

PROPERNESS OF DEFINABLE FORCING NEED NOT BE PROJECTIVE

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ABSTRACT. We give a negative consistency result for the projective definability of properness when definability is required of the domain and ordering of a forcing notion. Starting in L , we construct a uniformly closed family of partial orders and a forcing extension with no new countable sequences of ordinals in which the set of parameters giving proper partial orders does not belong to $L(\mathbb{R})$. The construction uses coanalytic incompatibility graphs on codes for club-shooting conditions. Finite independent sets in these graphs give analytic presentations, which can be replaced by closed presentations without changing the forcing. An almost-disjoint coding refinement gives the same nonprojectivity conclusion for closed presentations whose generic filter is recovered from a single generating real by a fixed Borel relation.

1. INTRODUCTION

Zapletal asks whether properness of definable, for example analytic, forcing notions is a projective property [5, Question 2.1.4]. We give a negative consistency answer when definability is required of the domain and ordering alone. Analyticity of these two relations does not imply analyticity of incompatibility, which is also required for Suslin forcing in the sense of [3, Definition 3.2].

A *closed presentation* is a partial order $(P, \leq)$ such that P is a closed subset of a fixed zero-dimensional Polish space and $\leq$ is closed in the square of that space. All the spaces below can be identified with closed subspaces of ω^ω . The word “closed” here is topological; it does not mean countably closed as a forcing. We do not assume that a presentation is separative. Zapletal likewise permits nonseparative presentations in the discussion preceding his question [5, Chapter 2, p. 5].

A family $(\mathbb{P}_a : a \in \omega^\omega)$ is *uniformly closed* if there are closed sets $D \subseteq \omega^\omega \times \omega^\omega$ and $O \subseteq \omega^\omega \times \omega^\omega \times \omega^\omega$ such that $\mathbb{P}_a = (D_a, O_a)$, with O_a a partial ordering of the nonempty set D_a . We regard a as a parameter for a fixed definition. Equivalently, we may pass continuously from a to a pair of ordinary closed-set codes for D_a and O_a .

We use the standard definition of properness. If θ is sufficiently large, a condition $q \leq p$ is an *M-master condition*, for a countable elementary submodel $M \prec H_\theta$ with $P, p \in M$, if

$$q \Vdash \dot{G} \cap M \text{ is } (P \cap M)\text{-generic over } M.$$

Equivalently, for every dense set $D \subseteq P$ with $D \in M$, the set $D \cap M$ is predense below q . The forcing P is *proper* if, for every sufficiently large regular θ , every countable $M \prec H_\theta$ with $P \in M$, and every $p \in P \cap M$, there is an M -master condition $q \leq p$.

Theorem 1.1. *Assume $V = L$. There are a uniformly closed family $(\mathbb{P}_a : a \in \omega^\omega)$ and a forcing $\mathbb{S}$ adding no countable sequences of ordinals such that, in every $\mathbb{S}$ -generic extension W ,*

$$B^W = \{a \in \omega^\omega : W \models \mathbb{P}_a \text{ is proper}\} \notin L(\mathbb{R})^W.$$

The closed sets defining the family belong to L and are interpreted from the same codes in W . In particular, $W \models \text{CH}$, and B^W is not projective.

Corollary 1.2. *Relative to ZFC, it is consistent with ZFC + CH that the set of real codes for proper partial orders with closed domain and closed ordering relation is not projective. In that model the code set does not belong to $L(\mathbb{R})$.*

The construction begins with canonical disjoint stationary subsets of ω_1 in L . We give closed presentations of the forcings shooting clubs through their complements, then force without adding countable sequences of ordinals to change exactly which selected stationary sets survive. The presentations remain unchanged, while their properness recovers a new subset of ω_1 from a constructible sequence of parameters. This yields the nonmembership in $L(\mathbb{R})$.

The selected family members used for this coding add no reals and hence are not real forcings in the sense of [5, Definition 2.1.5]. An almost-disjoint coding refinement gives each family member a single generating real, with Borel recovery of its generic filter, while preserving the same properness pattern.

Theorem 1.3. *Relative to ZFC, it is consistent with ZFC + CH that there is a uniformly closed family $(\mathbb{R}_a : a \in \omega^\omega)$ with the following properties. Each $\mathbb{R}_a$ adds a real z generating the entire forcing extension, and a fixed Borel relation $\chi(p, z)$, in the chosen coding of conditions, recovers its generic filter. For every nonempty proper basic clopen set $O \subseteq 2^\omega$ there are conditions forming a two-element maximal antichain whose recovery relations are exactly $z \in O$ and $z \notin O$. Nevertheless,*

$$\{a : \mathbb{R}_a \text{ is proper}\} \notin L(\mathbb{R}).$$

In particular this set is not projective.

The recovery and basic-open-set requirements are those in Zapletal's definition of a real forcing [5, Definition 2.1.5]. The proof of Theorem 1.1 and Corollary 1.2 does not use the refinement, which is developed in Section 5.

Our presentations are nonseparative, and the construction does not establish analyticity of incompatibility. It does not resolve the versions requiring separative or Suslin presentations. We also make no assertion under additional large-cardinal hypotheses; in particular, the result does not contradict Zapletal's equivalence, under ZFC + CH + LC, between properness and preservation of ω_1 for projective partial orders [5, Lemma 2.1.3].

We use standard forcing and constructibility facts as in [1], and the closed-projection characterization of analytic sets as in [2, Chapter III].

2. PRESENTATIONS AND FINITE INDEPENDENT SETS

We order forcing notions so that smaller conditions are stronger. Preorders will be used as intermediate presentations. Compatibility and forcing equivalence always refer to their separative quotients.

Lemma 2.1. *Suppose $\pi : P \rightarrow Q$ is a surjective, order-preserving map between preorders. Assume that p and p' are compatible exactly when $\pi(p)$ and $\pi(p')$ are compatible, and that whenever $q \leq \pi(p)$ there is $p' \leq p$ with $\pi(p') \leq q$. Then π induces an isomorphism of separative quotients. In particular, P and Q are forcing-equivalent.*

Proof. The hypotheses give $p \leq^* p'$ if and only if $\pi(p) \leq^* \pi(p')$: order preservation and compatibility reflection give the reverse implication, while the lifting property lifts any witness to $\pi(p) \not\leq^* \pi(p')$. Surjectivity then gives the isomorphism. $\square$

For a Polish space X , let $\text{Fin}(X) = \coprod_{k < \omega} X^k$, with the disjoint-union topology. We use finite sequences rather than a space of unordered finite sets. Repetitions and the order of entries are immaterial to the forcing. For $u \in \text{Fin}(X)$ write $\text{supp}(u) = \text{ran}(u)$, and put $u \preceq v$ if

$\text{supp}(u) \supseteq \text{supp}(v)$. This is a closed preorder on $\text{Fin}(X)$. On each pair of components it is a finite intersection of finite unions of equality conditions.

Let $E \subseteq X^2$ be a symmetric irreflexive coanalytic relation. Define

$$P(E) = \{u \in \text{Fin}(X) : (\forall x, y \in \text{supp}(u)) \neg E(x, y)\},$$

ordered by $\preceq$. The domain is analytic, and two conditions are compatible exactly when the concatenation of their sequences belongs to $P(E)$. These assertions hold uniformly for coanalytic families of graphs.

Lemma 2.2. *Let Q be a partial order with a weakest condition 1_Q . Suppose every nonempty finite pairwise compatible subset of $Q \setminus \{1_Q\}$ has a strongest member. Let $V_0 \subseteq X$ and let $d : V_0 \rightarrow Q \setminus \{1_Q\}$ be onto. Suppose*

$$E(x, y) \iff x, y \in V_0 \text{ and } d(x) \perp_Q d(y)$$

is coanalytic. Then $P(E)$ is forcing-equivalent to Q .

Proof. Map $u \in P(E)$ to the strongest member of $\{d(x) : x \in \text{supp}(u) \cap V_0\}$, or to 1_Q if this set is empty. This map π is order-preserving and onto. Two sequences are compatible exactly when all pairs in their union are E -independent, equivalently when $\pi(u)$ and $\pi(v)$ are compatible. Given $q \leq \pi(u)$ with $q \neq 1_Q$, append a code for q to u ; this lifts q to an extension of u . The case $q = 1_Q$ is immediate. Apply Lemma 2.1. $\square$

The points outside V_0 are not discarded. They are isolated vertices of E and impose no restriction on the separative quotient. This is why the validity of a code need not be an analytic condition on the domain of the resulting forcing.

Lemma 2.3. *Let I, Y be zero-dimensional Polish spaces, let $A \subseteq I \times Y$ be analytic with nonempty sections, and let $\preceq$ be a closed preorder on Y . There is a uniformly closed family of partial orders $(\hat{P}_a : a \in I)$ such that $\hat{P}_a$ is forcing-equivalent to $(A_a, \preceq)$ for every a .*

Proof. Choose a closed set $F \subseteq I \times Y \times \omega^\omega$ whose projection onto $I \times Y$ is A . A condition of $\hat{P}_a$ is (u, w, k) with $(a, u, w) \in F$ and $k < \omega$. Put

$$(u, w, k) \leq (v, z, l) \iff (u, w, k) = (v, z, l) \text{ or } (u \preceq v \text{ and } k > l).$$

The domain is closed. The ordering, including its domain requirements, is a finite union of closed sets. It is reflexive and transitive, and the strict inequality of the integer coordinates gives antisymmetry.

Projection to u is onto A_a and order-preserving. A common extension in A_a lifts to a common extension in $\hat{P}_a$ by choosing a closed-set witness and a sufficiently large integer coordinate. The same observation gives the lifting property for an extension of one projected condition. Compatibility in the other direction follows by projection. Apply Lemma 2.1.

Products and countable disjoint unions used here admit fixed closed embeddings into ω^ω . After making these identifications, the domains and ordering relations are jointly closed in the parameter and condition coordinates. $\square$

3. CONSTRUCTIBLE STATIONARY SETS AND COANALYTIC CODES

Work in L , and put $\kappa = \omega_1^L$. Use a canonical well-order $<_L$ which orders sets by their first constructibility stage before ordering the sets appearing at one stage. Its restrictions to the relevant initial segments are absolute. For every infinite $\beta < \kappa$, let $e_\beta : \omega \rightarrow \beta$ be the $<_L$ -least bijection. For $\xi < \kappa$ and $n < \omega$ define

$$S_{\xi, n} = \{\beta < \kappa : \beta > \max\{\xi, \omega\} \text{ and } e_\beta(n) = \xi\}.$$

Lemma 3.1. *There are $n_* < \omega$ and distinct ordinals ξ_*, ξ_i for $i < \kappa$ such that $S_* = S_{\xi_*, n_*}$ and $S_i = S_{\xi_i, n_*}$ are stationary. All these stationary sets are pairwise disjoint.*

Proof. For each ξ , the union of the sets $S_{\xi, n}$ contains a tail of κ , so some $S_{\xi, n}$ is stationary. One value n_* works for κ many ξ ; choose ξ_* and the ξ_i among them. Disjointness follows because $e_\beta(n_*)$ has only one value. $\square$

The canonical enumerations ensure definability of the individual sets; the stationary indices are selected in L .

For the model codes below, fix $\mathbf{T} = \mathbf{ZFC}^- + V = L + \text{“every set is countable”}$, where $\mathbf{ZFC}^-$ omits Power Set and includes full Separation, Collection, and Choice. Recall that $H_\kappa^L = L_\kappa \models \mathbf{T}$. Every finite tuple from this structure is fixed by the collapse of a countable elementary submodel, since its entries are hereditarily countable. Every transitive model of $\mathbf{T}$ is an L_α and computes e_β correctly for each infinite ordinal β it contains: it contains a bijection $\omega \rightarrow \beta$, and the stage-respecting order $<_L$ ensures that no earlier candidate is omitted.

For $T \subseteq \kappa$, let $\text{CU}(T)$ consist of the countable closed bounded subsets of T , including the empty set, ordered by end-extension. The empty set is the weakest condition. For nonempty conditions,

$$c \leq d \iff c \cap (\max(d) + 1) = d.$$

Two nonempty conditions are compatible exactly when one end-extends the other. Therefore a nonempty finite pairwise compatible family has a strongest member.

We now give a uniform coanalytic graph representing incompatibility for $\text{CU}(\kappa \setminus S_{\xi, n})$. The parameter is a pair $a = (n, r)$, where r is a real coding a well-order of type ξ . We use a fixed coding of binary relations on subsets of ω ; arbitrary reals are allowed as parameters, whether or not they code well-orders. The parameter space is identified with ω^ω .

Let $X = \omega^\omega$ be a space of raw codes for countable relational structures with finitely many distinguished elements. For a code x , write M_x for its coded structure. Say that $\text{Valid}(a, x)$ holds if M_x is well-founded and extensional, satisfies $\mathbf{T}$, and has correctly distinguished elements for the parameter r , an ordinal ξ , a nonempty set c , and its maximum b . Require in addition that M_x verifies that r is isomorphic to the ordinal ξ and that c is a closed bounded set of ordinals, with maximum b , avoiding the set given by the formula defining $S_{\xi, n}$.

Agreement of the distinguished real with the actual parameter r is required digit by digit and is Borel. Satisfaction of $\mathbf{T}$ and the remaining first-order assertions is also Borel, so Valid is coanalytic, with external well-foundedness supplying the only additional complexity. The internally required isomorphism from r to ξ becomes an actual isomorphism after collapse; hence an ill-founded parameter has no valid codes.

For a valid x , let c_x be the collapse of its distinguished c . The preceding absoluteness facts give $c_x \in \text{CU}(\kappa \setminus S_{\xi, n}) \setminus \{\emptyset\}$. Conversely, every nonempty condition has a valid code: take a countable elementary submodel of L_κ containing r and the condition. Its collapse fixes both.

Lemma 3.2. *There is a coanalytic relation $E(a, x, y)$ whose sections E_a are symmetric irreflexive graphs and which, for valid parameters $a = (n, r)$ in L , satisfies*

$$E_a(x, y) \iff \text{Valid}(a, x) \wedge \text{Valid}(a, y) \wedge c_x \perp_{\text{CU}(\kappa \setminus S_{\xi, n})} c_y.$$

Invalid codes are isolated vertices of E_a .

Proof. From a raw code x , extract the coded relation on the elements which are predecessors of its distinguished maximum b , together with b itself. Mark those elements which belong to its distinguished c . For a valid code this is a well-order of type $b + 1$, marked exactly by c_x .

Let $C(x, y)$ assert that there is an isomorphism from the marked order extracted from y onto a downward-closed initial segment of the marked order extracted from x . Preservation of the

markings is required in both directions on that initial segment. A candidate isomorphism is coded by a real, and all its required properties are Borel. Thus C is analytic on all raw codes. For valid codes it asserts precisely that c_x end-extends c_y .

Define

$$E(a, x, y) \iff \text{Valid}(a, x) \wedge \text{Valid}(a, y) \wedge \neg C(x, y) \wedge \neg C(y, x).$$

This is coanalytic and symmetric. It is irreflexive because the identity isomorphism witnesses $C(x, x)$ for every valid code. The stated interpretation on valid codes follows from the compatibility criterion for end-extension. $\square$

Proposition 3.3. *There is a uniformly closed family $(\mathbb{P}_a : a \in \omega^\omega)$ such that, for every well-order code $r \in L$ of type $\xi < \kappa$ and every $n < \omega$,*

$$\mathbb{P}_{(n,r)} \simeq \text{CU}(\kappa \setminus S_{\xi,n})$$

in L . The same forcing equivalence holds in every extension of L adding no countable sequences of ordinals, where the club-shooting forcing is computed in that extension.

Proof. Use the coanalytic graph of Lemma 3.2. The set of pairs (a, u) with $u \in P(E_a)$ is analytic. Its ordering is the restriction of the fixed closed support-inclusion preorder on $\text{Fin}(X)$. Apply Lemma 2.3 to obtain the uniformly closed family $\mathbb{P}_a$. All sections are nonempty, since the empty sequence belongs to $P(E_a)$, including for invalid parameters.

For a valid parameter in L , the decoding map from its valid codes is onto the nonempty club-shooting conditions. These conditions satisfy the hypothesis about finite compatible families in Lemma 2.2. That lemma gives the required forcing equivalence.

In an extension W adding no countable sequences of ordinals, the reals, ω_1 , code interpretations, and countable subsets of κ are unchanged. Thus the closed presentations and the club-shooting posets are unchanged, and the same maps still witness the forcing equivalences. $\square$

4. CHANGING PROPERNESS WITHOUT ADDING COUNTABLE SEQUENCES

We use the following standard club-shooting facts; see also [4, Section 2.1] for more general distributivity results.

Lemma 4.1. *Let $T \subseteq \omega_1$ be stationary. Then $\text{CU}(T)$ adds a club subset of T , adds no countable sequences of ordinals, and preserves every stationary subset of T . Moreover,*

$$\text{CU}(T) \text{ is proper} \iff T \text{ contains a club.}$$

Proof. Given a condition p and a name $\dot{f}$ for a countable sequence of ordinals, choose a countable $N \prec H_\theta$ containing the parameters, with $\delta = N \cap \omega_1 \in T$. Build a descending sequence in N below p , deciding $\dot{f}(n)$ at step n and with maxima cofinal in δ . Its union together with δ is a lower bound deciding $\dot{f}$. Thus no countable sequences of ordinals are added. The generic union is a club in T , since conditions with arbitrarily large maxima are dense and end-extension preserves closedness below each maximum.

For a stationary $S \subseteq T$ and a name $\dot{D}$ for a club, choose instead $\delta \in S$ and arrange that the sequence also forces points of $\dot{D}$ cofinal in δ . The lower bound forces $\delta \in S \cap \dot{D}$, proving preservation of S .

If $C \subseteq T$ is club, conditions with maximum in C form a dense countably closed suborder: a descending sequence has a lower bound obtained by adjoining the supremum of its maxima to its union. Otherwise, the generic club destroys the stationary complement of T , so $\text{CU}(T)$ is not proper. $\square$

Proof of Theorem 1.1. Start in L and choose the family from Proposition 3.3. Choose n_*, S_*, S_i and the corresponding ordinals as in Lemma 3.1. In L fix real codes r_i of the ordinals ξ_i , and put $a_i = (n_*, r_i)$. The sequence $\langle a_i : i < \kappa \rangle$ belongs to L .

First force with countable partial functions from κ to 2, ordered by extension, obtaining a new set $A \subseteq \kappa$. Countable closure preserves κ and the stationary sets S_i, S_* . In $L[A]$ put

$$T = \kappa \setminus \bigcup_{i \in A} S_i.$$

Since $S_* \subseteq T$, the set T is stationary. Force next with $\text{CU}(T)$, and let $C \subseteq T$ be the resulting club. Write $W = L[A][C]$, and let $\mathbb{S}$ be the two-step iteration just described.

By countable closure and Lemma 4.1, neither step adds countable sequences of ordinals. Hence

$$\mathbb{R}^W = \mathbb{R}^L, \quad \omega_1^W = \kappa, \quad W \models \text{CH}.$$

If $i \in A$, then $C \subseteq \kappa \setminus S_i$. By Lemma 4.1, $\text{CU}(\kappa \setminus S_i)$ is proper in W . If $i \notin A$, disjointness gives $S_i \subseteq T$, so Lemma 4.1 preserves its stationarity in W . Therefore $\text{CU}(\kappa \setminus S_i)$ is not proper in W . In either case $\kappa \setminus S_i$ remains stationary, since it contains the preserved stationary set S_* . By Proposition 3.3, the closed presentation still represents this club-shooting forcing. Thus

$$(1) \quad W \models \mathbb{P}_{a_i} \text{ is proper} \iff i \in A.$$

Each selected forcing still adds no countable sequences of ordinals.

Since W and L have the same reals, $L(\mathbb{R})^W = L$. If B^W belonged to L , the constructible parameter sequence and (1) would recover $A = \{i < \kappa : a_i \in B^W\}$ in L , contradicting that A is new. Thus $B^W \notin L(\mathbb{R})^W$, and hence B^W is not projective. $\square$

Proof of Corollary 1.2. Work in the extension from Theorem 1.1. The jointly closed family gives a continuous map $a \mapsto c(a)$, coded in L , to valid closed-presentation codes: in a tree code, the section at a is obtained by substituting $a \upharpoonright k$ at level k . The preimage of the properness-code set under c is B^W , so that code set cannot belong to $L(\mathbb{R})^W = L$ and in particular cannot be projective. The construction is a set-forcing argument over L , giving the stated relative consistency. $\square$

5. A SINGLE-REAL REFINEMENT

We prove Theorem 1.3 by appending almost-disjoint coding to the forcing of finite independent sets in a coanalytic graph. The main point is to retain an analytic presentation with Borel recovery of the generic filter before applying Lemma 2.3.

5.1. Almost-disjoint coding. Let E be a coanalytic symmetric irreflexive relation on $X = \omega^\omega$. If G_0 is $P(E)$ -generic, its union $H = \bigcup \{\text{supp}(u) : u \in G_0\}$ is independent and

$$G_0 = \{u \in P(E) : \text{supp}(u) \subseteq H\},$$

by directedness of G_0 . Here $H \subseteq X^V$: throughout this construction, X and E in forcing extensions denote the ground-model sets, even if $P(E)$ adds reals.

Fix a bijection between $\omega^{<\omega}$ and ω , and put

$$a_x = \{\ulcorner x \upharpoonright n \urcorner : n < \omega\} \quad (x \in X).$$

This is a continuous almost-disjoint family of infinite sets; every finite union of its members has infinite complement. Let $\mathcal{C}$ be the countable Boolean algebra of clopen subsets of 2^ω , with discrete codes. In $V[H]$, for finite $F \subseteq H \times \omega$, put

$$Z_F = \{z \in 2^\omega : (\forall (x, k) \in F)(\forall m \in a_x)[m \geq k \implies z(m) = 0]\}.$$

Conditions of $\mathbb{A}(H)$ are pairs (C, F) with $C \in \mathcal{C} \setminus \{\emptyset\}$ and $C \cap Z_F \neq \emptyset$, ordered by shrinking C and enlarging F . Consistency is a Borel finite test: a basic cylinder $[s]$ meets Z_F exactly when none of the restraints forbids a 1 specified by s , and every clopen set is a finite union of cylinders.

Lemma 5.1. *$\mathbb{A}(H)$ has a dense σ -centered suborder and adds a new real $z \in 2^\omega$ such that, for every $x \in X^V$,*

$$(2) \quad x \in H \iff |\text{supp}(z) \cap a_x| < \omega,$$

where $\text{supp}(z) = \{m : z(m) = 1\}$. Its generic filter is

$$(3) \quad G_1 = \{(C, F) \in \mathbb{A}(H) : z \in C \cap Z_F\},$$

and $V[G_0][G_1] = V[z]$.

Proof. Conditions with a basic cylinder as first coordinate are dense. For fixed s , finitely many conditions $([s], F)$ have a common extension obtained by taking the union of their restraints, with consistency witnessed by extending s by zeros. This proves σ -centeredness. Refining to cylinders of arbitrary length produces a unique real z satisfying every clopen requirement and restraint in the generic filter.

For $x \in H$, it is dense to add a restraint (x, k) , choosing k above the current stem. For $x \in X^V \setminus H$, almost disjointness allows a 1 to be placed arbitrarily far along a_x outside the finitely many restrained branches. These density arguments prove (2). The infinite complement of finitely many branches likewise allows an extension disagreeing with any prescribed real in $V[H]$, so z is new.

If $z \in C \cap Z_F$, then (C, F) is compatible with every $(D, J) \in G_1$: the pair $(C \cap D, F \cup J)$ is consistent, as witnessed by z . The finite consistency test is absolute to $V[H]$, so genericity gives $(C, F) \in G_1$. This proves (3); the other inclusion follows from the choice of z . The two equations recover H , hence G_0 , and G_1 from z , proving $V[G_0][G_1] = V[z]$. $\square$

5.2. Presentation and recovery. In V , let $R(E)$ consist of triples $p = (u, C, F)$, where $u \in P(E)$, $C \in \mathcal{C} \setminus \{\emptyset\}$, and $F \subseteq \text{supp}(u) \times \omega$ is finite with $C \cap Z_F \neq \emptyset$. As before, finite sets are represented by finite sequences and inclusion refers to their supports. Put

$$(v, D, J) \leq (u, C, F) \iff \text{supp}(v) \supseteq \text{supp}(u), \quad D \subseteq C, \quad J \supseteq F.$$

The domain is analytic: independence is analytic, incidence of restraints with $\text{supp}(u)$ is closed, and consistency is Borel. The ordering is the restriction of a closed ambient preorder, since finite support inclusion is closed and clopen codes are discrete. These assertions are uniform in coanalytic families E .

The map $(u, C, F) \mapsto (u, (C, F))$ preserves and reflects order and has dense image in $P(E) * \dot{\mathbb{A}}(\dot{H})$. Indeed, strengthen the first coordinate of an iteration condition to decide its finite second-coordinate data. All those data belong to ground-model sets, including the vertices in X^V . The first coordinate then forces the vertices of F into $\dot{H}$, so adjoining them to its independent sequence gives a condition in the image. Thus $R(E)$ presents this iteration.

For any raw triple $p = (u, C, F)$ define the Borel relation

$$(4) \quad \begin{aligned} \chi_0(p, z) &\iff z \in C, \\ &(\forall x \in \text{supp}(u)) \quad |\text{supp}(z) \cap a_x| < \omega, \\ &(\forall (x, k) \in F) \quad \text{supp}(z) \cap a_x \subseteq k. \end{aligned}$$

Proposition 5.2. *The forcing $R(E)$ has a closed presentation whose extension is generated by a new real z and whose generic filter is recovered by a Borel relation χ . For every nonempty proper clopen set $O \subseteq 2^\omega$, two conditions form a maximal antichain whose recovery relations are exactly $z \in O$ and $z \notin O$. The construction is uniform in coanalytic families of graphs.*

Proof. If G is $R(E)$ -generic, Lemma 5.1 gives

$$G = \{p \in R(E) : \chi_0(p, z)\}.$$

To check the reverse inclusion, suppose $\chi_0((u, C, F), z)$ and $(v, D, J) \in G$. By (2), both supports are contained in H , so their union is independent. Also z witnesses the consistency of $(C \cap D, F \cup J)$. The finite test gives a ground-model common extension, and genericity yields $(u, C, F) \in G$.

The conditions $(\emptyset, O, \emptyset)$ and $(\emptyset, 2^\omega \setminus O, \emptyset)$ are incompatible. Every condition can be strengthened by intersecting its clopen coordinate with one of these two sets, so they form a maximal antichain. Their recovery formulas reduce to the asserted membership relations.

Apply Lemma 2.3 and write its conditions as (p, w, k) . Projection to p preserves compatibility and induces an isomorphism of separative quotients. If $\hat{G}$ is generic and G is its projected generic filter, every lift of a member of G is compatible with every member of $\hat{G}$ and hence belongs to $\hat{G}$. Thus $\chi((p, w, k), z) = \chi_0(p, z)$ recovers $\hat{G}$ exactly, and arbitrary lifts of the two displayed conditions remain a maximal antichain. The generating-real assertion follows from Lemma 5.1. Finally, z has a Borel name: for its m th bit, use the conditions whose clopen coordinate is contained in $\{z : z(m) = 1\}$; cylinders deciding longer initial segments are dense. $\square$

Proof of Theorem 1.3. Use the graphs E_a from Lemma 3.2 and the extension W from Theorem 1.1, with the new set $A \subseteq \kappa$ coded there. Form $R(E_a)$ and its closed presentation $\mathbb{R}_a$ uniformly, with codes in L . Since W adds no reals, these presentations and their Borel recovery relation have the same interpretation in L and W .

Recall that $\kappa = \omega_1^L = \omega_1^W$, that $\langle a_i : i < \kappa \rangle \in L$, and that $P(E_{a_i}) \simeq \text{CU}(\kappa \setminus S_i)$. For $i \in A$, this forcing is proper, and its iteration with the σ -centered coding forcing is proper by the two-step iteration theorem [1]. For $i \notin A$, it preserves κ but adds a club disjoint from the stationary set S_i . The ccc coding forcing preserves κ and this club, so the iteration still destroys the stationarity of S_i and is not proper. Hence

$$W \models \mathbb{R}_{a_i} \text{ is proper} \iff i \in A.$$

The generating-real, recovery, and antichain requirements follow from Proposition 5.2. As in Theorem 1.1, the properness set recovers $A \notin L$ from the constructible parameter sequence, so it does not belong to $L(\mathbb{R})^W = L$ and is not projective. $\square$

6. AI ACKNOWLEDGEMENT

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